

University of Neuchâtel
Faculty of Science
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Micromammals and mountain habitats

*Effects of land use, local habitat structures and climatic gradients on
species abundance*

Master's Thesis

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Abstract

Mountain agricultural landscapes are composed of grasslands, wetlands, forest, edges, aquatic structures and other local microhabitats that may influence micromammal communities in different ways. This thesis examines how land-use typologies, local habitat structures and climatic gradients are associated with the abundance of small mammals in Swiss mountain landscapes. The analysis was based on capture data collected by Faune Concept over the last ten years in the Swiss Alps, with 2684 conserved records, 504 sampling stations and 11 species. For each species, the most frequent typologies, structures and typology x structure combinations were first described. Their statistical association with abundance was then tested using negative binomial models. Climatic covariates (BIO1, BIO2, BIO5, BIO12 and PCA1) were introduced separately to assess the robustness of habitat effects.

The results show that habitat associations are strongly species-specific. Wetlands and aquatic structures were positively associated with *Microtus agrestis*, *Microtus arvalis* and *Neomys fodiens*, but negatively associated with *Clethrionomys glareolus* and *Chionomys nivalis*. Forest habitats and forest edges were especially important for *Clethrionomys glareolus*, while *Sorex araneus* showed a negative association with forest habitats. Most robust effects remained significant after climatic adjustment, suggesting that the observed patterns were not only due to climatic variables.

Overall, the study highlights the importance of maintaining heterogeneous mountain landscapes, where wetlands, aquatic structures and forest edges provide complementary ecological contexts for different species.

Keywords: micromammals; mountain agriculture; land use; habitat structures; wetlands; forest edges.

Table of Contents

1. Introduction.....	1
2. Literature review.....	3
2.1. Land use, forest habitats, and edge structures	4
2.2. Open, wetland, and aquatic habitats	4
2.3. Influence of climate.....	5
3. Materials and Methods.....	6
3.1. Study area	6
3.2. Data used and unit of analysis	6
3.3. Dataset structure	8
3.4. Climatic variables used.....	8
3.5. Data harmonization and hierarchical analysis	9
3.5.1. Harmonization of the different categories	9
3.5.2. A second targeted hierarchical pass	10
3.6. Statistical analyses.....	10
4. Results.....	12
4.1. <i>Apodemus alpicola</i>	12
4.2. <i>Apodemus flavicollis</i>	12
4.3. <i>Apodemus sylvaticus</i>	13
4.4. <i>Chionomys nivalis</i>	13
4.5. <i>Clethrionomys glareolus</i>	14
4.6. <i>Microtus agrestis</i>	14
4.7. <i>Microtus arvalis</i>	15
4.8. <i>Microtus lavernedii</i>	15
4.9. <i>Neomys fodiens</i>	16
4.10. <i>Sorex antinorii</i>	17
4.11. <i>Sorex araneus</i>	17
4.12. General synthesis.....	18
5. Discussion.....	21
5.1. Contrast between wetlands, forests, and grasslands	21
5.2. The contribution of local structures	22
5.3. What does the second hierarchical pass teach us?	23
5.4. Species with no robust signal: no effect or no statistical power?	24
5.5. Limitations of the analysis.....	24
5.6. Ecological and management implications	25
5.7. A review of the assumptions made in the introduction	25
6. Conclusion	27
Bibliography	28
Appendix.....	31

List of Figures

Figure 1. Swiss cantons included in the micromammal surveys	6
Figure 2. Sampling coverage and total abundance by species	7
Figure 3. Extract of the worked dataset used for subsequent analyses	8
Figure 4. Effect sizes of significant habitat associations	18
Figure 5. Species-specific responses to harmonized habitat categories.....	19
Figure 6. Robustness of significant habitat effects after climatic adjustment.....	20

List of Tables

Table 1. Harmonized typology and structure categories used in the analysis.....	9
Table 2. Summary of the second hierarchical pass	24
Table 3. Climate-adjusted robustness of significant effects	34

1. Introduction

The biodiversity in the world is facing a major decline during the last decades, mainly linked to land use change due to human activity (IPBES,2019; WWF France, 2024). According to the WWF and his report “living planet”, over the past fifty years, there has been an average decline of 73% in the size of 34,836 wild animal populations across 5,495 species (WWF France,2024). Changes in land uses are one of the main reasons of the reorganization of the communities of small mammals, because they modify simultaneously the structure of the vegetation, the availability of resources, the exposition to predators and the connectivity of habitats (Paniccia and al., 2022; Torre et al., 2024).

Micromammals form a relevant group to study the connexions between habitats and abundance, due to their quick life cycle, their strong dependency to the microstructures, to the variation of the vegetation cover and climate. (Čepelka & Dokulilová, 2025). They occupy a key position in the terrestrial ecosystems, contributing to the food web, soil tillage, dispersing of seeds and controlling invertebrate populations (Kauer et al., 2025). In mountain agricultural landscapes, effects of land use on micromammals are not uniform (Kratschmer and al., 2024). Species do not only react to bigger categories of soil utilisation, but also to fine structures such as forest edges, hedges, small patch of forest, ponds or drystone walls (Schlinkert and al., 2016). Studies conducted in Europe show that permanent or semi natural habitats, wooden structures generally support more abundant or mor diversified communities than a simplified landscape (Broughton and al., 2014.) On the opposite, forest species and those from open environment or humid do not meet the same environmental standards, so that only one category of soil usage can favour some species and disadvantages some others, regarding their needs in terms of humidity, opening or closing environments (Balčiauskas et al., 2017).

Mountain landscapes are characterised by a wide variety of permanent meadow, hedges, edges, forests, humid areas, offering a wide range of micro-habitats crucial for numerous species (Marshall & Moonen, 2002). This is important to assess what uses of soil and what structures fine associated explain the abundance of micromammals in a landscape if we want to see them thrive (Dorigo et al., 2021). We should in this case think in terms of general typologies and ecological categories finer (Torre and al., 2024).

In this view, the objective of this study is to identify the types of typologies, local structures and the combinations between typologies and structure that are associated with the abundance

of micromammals species listed, while verifying if those effects persist when we consider the climatic gradients available in our data. To do that, we will analyse data from catches made the 10 last years in the Swiss Alps. Furthermore, in this study, we'll try to answer to the following assumptions: The first assumption is that wetlands and aquatic habitats are positively associated with the abundance of species associated with wetland environments, although these same habitats are not necessarily favourable to all species. The second hypothesis is that forest related habitats are positively associated with the abundance of species associated with forest environment, while certain species typically found in more open areas may be less abundant there. The third hypothesis is that the associations between typologies, local structures, and abundance remain detectable even after accounting for the climatic covariates BIO1, BIO2, BIO5, BIO12, and PCA1, suggesting that the observed effects do not solely reflect a regional climatic gradient.

2. Literature review

The micromammals are frequently used as biological indicators (Čepelka & Dokulilová, 2025). This is due to the composition of their communities that vary strongly with the structure of the habitats and the management of the environment, which allows to connect precisely the biodiversity observed to the characteristics of the studied area (Shilereyo and al., 2023). In agricultural landscapes, communities tend to simplify when the land use surfaces increase and that the natural areas decrease (Heroldová and al., 2007). On the opposite, micromammal communities become more varied when they are imbedded into diversified cultures, prairies, hedges, small woods and non-cultivated areas, which increase the diversity of resources and refuges for micromammals (Paniccia and al., 2022). For instance, a study centered around small forest patches, showed that the composition of the micro-habitats was able to explain an important part of the variations of abundance of *Apodemus agrarius* et *Apodemus sylvaticus* (Dorigo et al., 2021), which underline that micromammals interact with different level of spatial organisation and not only to a single dominant environment. This ecological sensibility can be explained by the fact that micromammals need precise combinations of humidity, vegetation cover and spatial continuity (Torre et al., 2022). Hence, small differences of local structures can product measurables changes in terms of abundance of specific richness (Benedek et al., 2021). For a problematic centered on the use of soil, it means that it is methodologically relevant not to limit to broad ecological categories, but to integrate too the fine ecological structures that influence the living conditions for many micromammals species (Dorigo et al., 2021).

2.1. Land use, forest habitats, and edge structures

Forest environments regularly host species and communities distinct to those present in open lands (Heroldová et al., 2007). Many studies show that wood patches, hedges and edges play a structuring role in the distribution of small mammals in the European agricultural landscapes (Schlinkert et al., 2016). Studies on hedges and forest edges show that small mammal specialist in forest tend to favour the wooden components connected to the forest matrix (Kratschmer et al., 2024.) However, changes of those environmental contexts can lead to changes in communities (Schlinkert et al., 2016). For example, land abandonment or the progressive closing of open spaces changes the species-specific richness in communities, with a decrease

of certain species linked to grassland or forest edges, benefiting species more adapted to more advanced forest steps (Balčiauskas et al., 2017).

2.2. Open, wetland, and aquatic habitats

Open grassy areas, vegetated margins and permanent, uncultivated habitats play a major role for several voles (Broughton and al., 2014). They furnish a stable cover, a continuous of food availability and usable refuges during the unfavourable agricultural period (Heroldová and al., 2007). Hence, higher abundance of voles was shown in stable grassy margins and habitats than inside the disturbed agricultural areas (Rodríguez-Pastor and al., 2016; Kauer et al., 2025). This confirms the importance of structures that acts as refuges for the population of micromammals. *Microtus agrestis* populations have been shown to decrease when the untouched grass areas were destroyed (CITE). Humid habitats represent a special case, because they gather a combination of humid soil, dense vegetation and a high number of invertebrates (Greenwood and al., 2002;). This type of habitat can accommodate rare species, which are missing in dryer habitats (van der Putten and al., 2025). *Neomys fodiens* is the best example, because its presence is highly connected to humid zones, rivers, streams, wetlands and sites with low and dense vegetation, whereas dry and open spaces are less favourable for this species (Van der Putten et al., 2025). Hence, maintaining a diversity of humid and open habitats seems important to have a diversified community of micromammals.

2.3. Influence of climate

One of a key aspect of the ecological analysis of micromammals consists of distinguish what is part of the effect of habitats, and what is part of the climate effects (Torre and al., 2024). Because temperature, temperature range and precipitation influence vegetation, food resources, and population dynamics (Deitloff and al., 2010), climatic factors will likely affect micromammals populations (CITE). In this regard, studies conducted in agroecosystems in north-est of Spain show that the different uses of soil explain often more the variance than the climate at small scale, but that an important part of the community structure is shared between climate and use of soil (Torre and al., 2024). For example, studies on *Clethrionomys glareolus* shows that hotter years and episodes of extreme rains can reduce the abundance of the species, which confirm that the climate variables should be considered as important co-variables explanatory, without, however, downplaying the importance of the structure of habitats (Huerta-Schliemann and al., 2025.)

3. Materials and Methods

3.1. Study area

The study was conducted in diverse regions and mountain agricultural landscapes of Switzerland, distributed along an altitudinal gradient with different alpine bioclimatic zones. The trapping area for animals goes from 450m to 2300m.

Swiss cantons included in the micromammal surveys

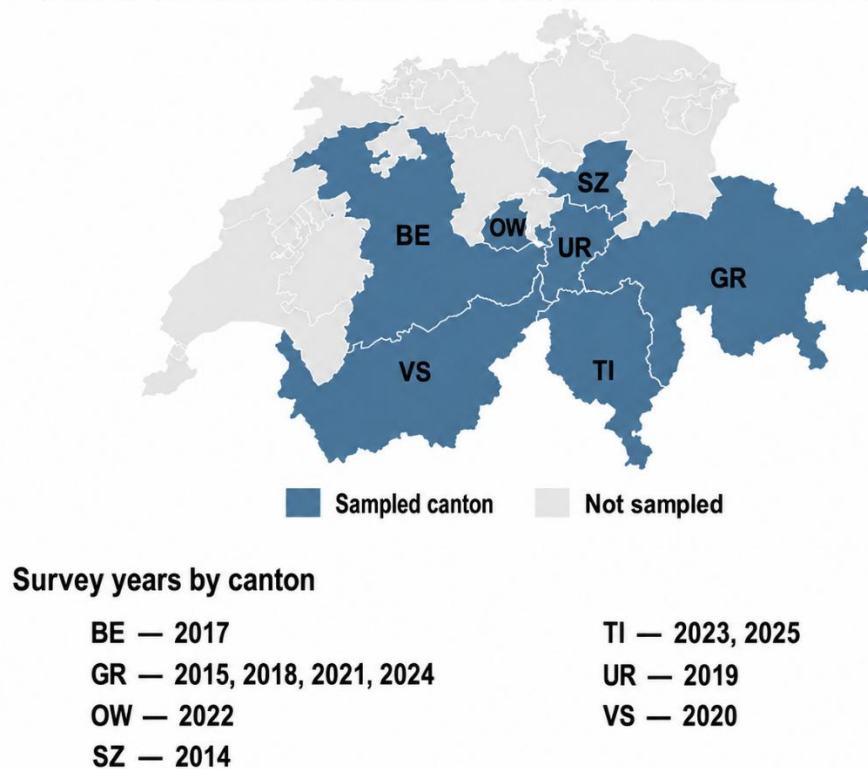


Figure 1. Swiss cantons included in the micromammal surveys

3.2. Data used and unit of analysis

The data analysed in this master's thesis were collected during field surveys conducted by Faune Concept. Faune Concept is a community of 5 ecology firms dedicated to the study of wildlife, in particular the mammals.

Gathered over the past ten years in several regions of the Swiss Alps, the data were not collected specifically for this project but were made available as part of a collaboration with the firm.

The data are the property of Faune Concept. Their use has been authorized for the purposes of this academic work. Field collection, initial identification of individuals are the responsibility of Faune Concept. The original dataset with all captures came in form of an Excel file, which listed all the captures of micromammals and the descriptive variables associated with the different stations. After sorting the data, the dataset consisted of 2684 lines (captures), corresponding to 504 stations and 11 micromammal species. Sampling coverage differed substantially among species, both in terms of occupied stations and total abundance (Figure 2).

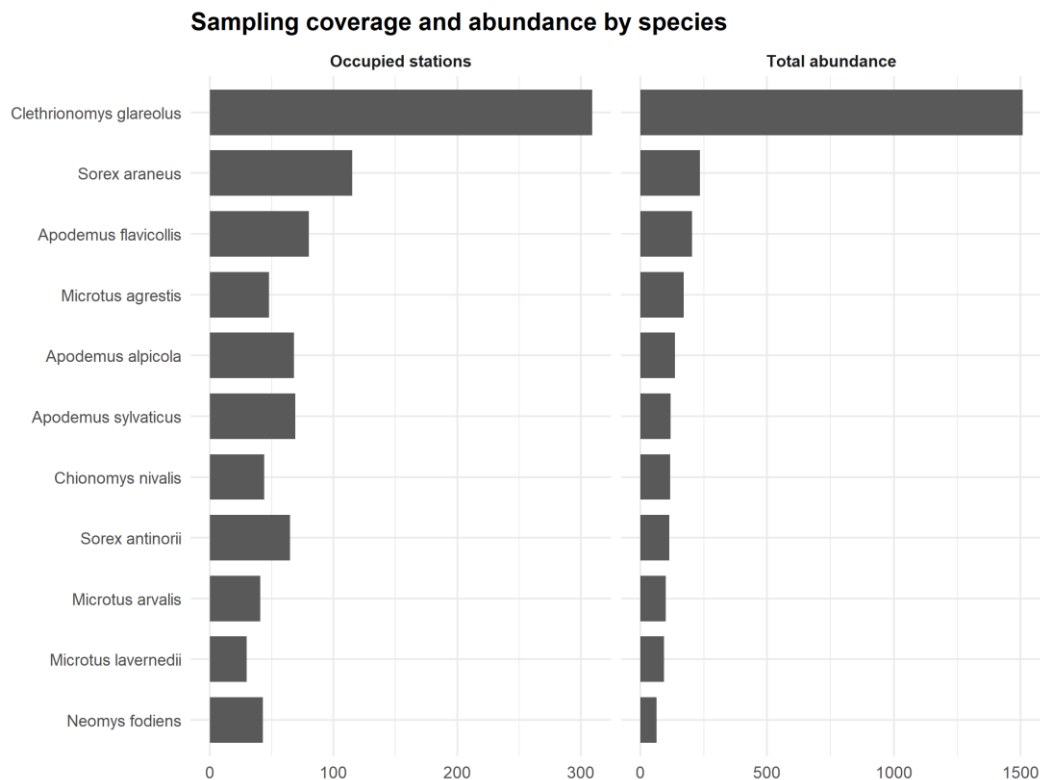


Figure 2. Sampling coverage and total abundance by species. The figure shows the number of occupied stations and the total abundance recorded for each species.

The typology describes the dominant environment surrounding the station at one hectare scale, and the structure describes the structural elements observed at the precise emplacement of the capture. A maximum of 2 structures per station could be indicated. The variables Structure 1 and structure 2 were considered as interchangeable and non-exclusive. A station could contribute to 2 structures simultaneously.

Some cells of the listing were empty regarding the typology and structure. They were therefore filed manually, based on the available data of the site map.geo.admin.ch. The following filters were used to determine in which type of environment and structure the stations were located:

- Standard land area statistics
- Wetland sites
- Raised bogs
- Flat bogs
- Dry grassland zones
- IFN forest diversity
- Land use areas
- Natural habitat map – Base
- Soils: Crop typology

3.3. Dataset structure

The final dataset for subsequent analyses was formatted into a new Excel file (Figure 3), in which, each line of the dataset corresponds to an observation of an individual captured, while columns represent: 1) the species observed, 2) the GPS coordinates of the location, 3) the type of dominant environment (typology), and 4) the main structures of the habitat (structure). Such information allows us to connect every observation with a specific ecological context and to analyse the relationship between the presence of a species and the characteristics of the habitat.

	ESPECE	CORCHONNEX	CORCHONNEY	TYPLOGIE	STRUCTURE	STRUCTURE 2	Elevation	lat (temperature)	lat2 (precipitation)	lat2 (temperature)	PC1		
1	Apodemus alpicola (Mulet alpicola)	650	593	132	504	Milieu forestiers (conifères)	Torrent (avec végétation)	1400.40	37.00	10701.00	90.00	185.00	0.01
2	Apodemus alpicola (Mulet alpicola)	702	628	169	097	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1308.00	31.00	12208.00	70.00	187.00	-0.18
3	Apodemus alpicola (Mulet alpicola)	702	628	169	097	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1308.00	31.00	12208.00	70.00	187.00	-0.18
4	Apodemus alpicola (Mulet alpicola)	702	628	169	097	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1308.00	31.00	12208.00	70.00	187.00	-0.18
5	Apodemus alpicola (Mulet alpicola)	745	608	158	092	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1461.00	25.00	12308.00	90.00	187.00	0.48
6	Apodemus fuscicola (Mulet à collier)	584	664	144	120	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1400.70	48.00	13389.00	94.00	185.00	-0.44
7	Apodemus fuscicola (Mulet à collier)	584	664	144	120	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1400.70	48.00	13389.00	94.00	185.00	-0.44
8	Apodemus fuscicola (Mulet à collier)	584	664	144	120	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1400.70	48.00	13389.00	94.00	185.00	-0.44
9	Apodemus fuscicola (Mulet à collier)	584	664	144	120	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1400.70	48.00	13389.00	94.00	185.00	-0.44
10	Apodemus fuscicola (Mulet à collier)	584	664	144	120	Milieu forestiers (conifères)	Ruisseau (avec végétation)	1400.70	48.00	13389.00	94.00	185.00	-0.44
11	Apodemus fuscicola (Mulet à collier)	652	598	189	754	Milieu forestiers (conifères)	Ruisseau (sans végétation)	1400.70	48.00	13389.00	94.00	185.00	-0.44
12	Apodemus fuscicola (Mulet à collier)	652	598	189	754	Milieu forestiers (conifères)	Ruisseau (sans végétation)	1400.70	48.00	13389.00	94.00	185.00	-0.44
13	Apodemus fuscicola (Mulet à collier)	659	153	135	607	Milieu forestiers (conifères)	Lisières forestières (conifères)	1117.00	36.00	17022.00	70.00	185.00	-0.88
14	Apodemus fuscicola (Mulet à collier)	659	153	135	607	Milieu forestiers (conifères)	Lisières forestières (conifères)	1489.20	41.00	12208.00	90.00	195.00	-0.21

Figure 3. Extract of the worked dataset used for subsequent analyses. The figure shows how the cells were filed in the excel field file.

3.4. Climatic variables used

In this study, climatic variables were used to assess the robustness of the relationships observed between habitat characteristics (typology and structure) and species abundance. This approach aims to distinguish the contributions of local ecological conditions and climatic gradients. A result that remains significant after climatic adjustment is interpreted as reflecting a relationship strongly linked to the typology and structure, whereas a result that disappears after

adjustment suggests that the observed association may be largely driven by the climatical conditions rather than by habitat characteristics.

Hence, the adjusted analyses integrated five climatic variables separately: bio1 (mean annual temperature), bio2 (mean diurnal temperature range), bio5 (maximum temperature of the warmest month), bio12 (annual precipitation), and PCA1 (the first principal axis of a principal component analysis summarizing the main multivariate climatic variables). The climate variables used in this study were retrieved from the Worldclim2 database, developed by Fick and Hijans (2017).

3.5. Data harmonization and hierarchical analysis

3.5.1. Harmonization of the different categories

A preliminary harmonization step was put in place to avoid directly comparing categories with different levels of precision (Table 1). The objective was to build a level of analysis consistent between stations, and to conserve more precise categories for complementary analyses. Hence, some categories were transformed as follows:

Variable	Harmonized category	Fine categories
Structure	Forest edges	Forest edges (coniferous); Forest edges (deciduous); Forest edges (mixed)
Structure	Built anthropogenic habitats	Built anthropogenic habitats
Structure	Aquatic habitats with vegetation	Stream (with vegetation); Torrent (with vegetation)
Structure	Aquatic habitats without vegetation	Stream (without vegetation); Torrent (without vegetation)
Structure	Rocky habitats	Anthropogenic rocky habitats; Natural rocky habitats
Structure	Semi-open woody structures	Scrub / shrub formations; Linear tree structures in agricultural landscapes
Typology	Forest habitats	Forest habitats (coniferous); Forest habitats(deciduous);Forest habitats (mixed); Open / disturbed forest habitats (deciduous)
Typology	Wetlands	Wetlands (lowland marshes); Wetlands (raised bogs); Wetlands (wet meadows)
Typology	Grasslands	Grasslands

Table 1. Harmonized typology and structure categories used in the analysis

As a complement, a second variable was created by merging “Aquatic habitats with vegetation” and “Aquatic habitats without vegetation” in a more general category: “Aquatic habitats”. The

variable was not replaced at the main level of analysis but has allowed to assess whether certain results were associated with a general aquatic structure or a more specific aquatic type.

3.5.2. A second targeted hierarchical pass

After the first harmonized analyse, a second hierarchical pass was done for the typologies and structures that had a visible effect after climatic adjustment. The goal was to test the detailed subcategories inside a broad, statistically significant category. Hence, for example, when a more general category such as “Wetlands” or “Forest habitats” was standing out in the harmonized analyses, the detailed subcategories were tested. The same logic was applied to the structures. However, this second pass was not conducted for the combinations of typology and structure together, because they were more numerous and more fragile from a statistical point of view.

3.6. Statistical analyses

All analyses were made using R, version 4.4.3, And RStudio, version 2026.01.2+418. Data preparation, analyses, table, and figure generation utilized the following packages: readxl, dplyr, tidyr, stringr, tibble, MASS, writexl, ggplot2, forcats, sf, rnaturalearth, and rnaturalearthdata.

At each level of analysis (Typologie ; Structure ; Typologie x structure), the three habitats categories most frequently occupied by the species were selected. Each category was then tested separately: sites where it was present were compared to all sites where it was absent. Abundances were analysed using generalized linear models with a negative binomial distribution and a logarithmic link function. When this model could not be used, a Poisson model was used.

For each association between a species and a category, an unadjusted model was first estimated. Five additional models were then constructed by separately adding one of the following climate variables: BIO1, BIO2, BIO5, BIO12, or PCA1. These variables were never included simultaneously in the same model.

Statistical significance was assessed using Wald tests. Model coefficients were transformed to obtain rate ratios (RR) and their 95% Wald confidence intervals (CI), which indicates the

uncertainty surrounding each RR estimate. A RR greater than 1 indicated a higher model-estimated abundance at sites containing the tested category, and an RR below 1 indicated a lower estimated abundance. Final significance was determined using Benjamini-Hochberg adjusted p-values (qBH). An association was considered significant when $qBH < 0.05$. All analyses were conducted separately for each species. Within each species, each selected habitat category was tested independently. The results were then divided into four categories. An effect was considered as robust if it was significant in the unadjusted model and remained significant after at least one climate adjustment. It was classified as limited to the raw model if it was significant only before adjustment. On the opposite, an effect was considered to be visible after adjustment if it became significant only after the introduction of a climate variable. All other cases were considered as “no effect detected.”

4. Results

For all results detailed below, Table 3 (Appendix) summarizes all statistically significant effects, while Figures 4, 5 and 6 illustrate effect sizes, species-specific responses and robustness after climatic adjustment.

4.1. *Apodemus alpicola*

With *Apodemus alpicola*, the three most frequent typologies are the grassland (51 stations, total abundance is 104), the forest habitats (12 stations, total abundance is 22) and the wetlands (5 stations, total abundance is 9). The 3 most frequent structures are the aquatic environment with vegetation (22 stations, total abundance is 50), the semi-open woody structures (17 sites, total abundance is 29) and forest edges (14 sites, total abundance is 34). The three most common combinations are Grasslands x Semi-open woody structures (17 sites, total abundance is 29), Grasslands x Aquatic habitats with vegetation (16 sites, total abundance is 42), and Grasslands x Forest edges (12 sites, total abundance is 30). From a statistical point of view, only the combination Grassland x aquatic environment with vegetation is significant. The model indicates an RR of 2.17 (95% CI: 1.06–4.46; qBH = 0.0395), which means that the observed abundance of the species is more than twice as high at sites with this combination. However, no robust effect was detected for the site types considered in isolation or for the structures alone. This effect persists after climate adjustment, indicating a relationship independent of the variables BIO1, BIO2, BIO5, BIO12, and PCA1.

4.2. *Apodemus flavicollis*

The three most used typologies for *Apodemus flavicollis* are Grasslands (50 sites, total abundance is 137), Wetlands (17 sites, total abundance is 31), and Forested habitats (12 sites, total abundance is 35). The three most common structures are Forest Edges (28 sites, total abundance is 69), Aquatic Environments with Vegetation (26 sites total abundance is 70), and Semi-open Woody Structures (17 sites total abundance is 44). The three most frequent combinations are Grasslands x Aquatic habitats with vegetation (18 stations, total abundance is 44), Grasslands x Forest edges (17 stations total abundance is 48), and Grasslands x Rocky habitats (14 stations total abundance is 34). No robust effect was detected in the raw model for this species. The most frequent categories therefore remain mainly descriptive. However, the

adjusted analyses show a negative association with Wetlands after adjustment for BIO1 and PCA1 (RR = 0.39; 95% CI: 0.20–0.76; qBH = 0.0179). This result should be interpreted as a visible effect after climatic adjustment rather than as a robust raw-model effect. It suggests that the relationship between *A. flavicollis* and wetlands may be partly revealed when climatic variation is considered.

4.3. *Apodemus sylvaticus*

The three most present typologies for *Apodemus sylvaticus* are Grasslands (47 sites, total abundance is 81), Wetlands (11 sites, total abundance is 26), and Forests (11 sites, total abundance is 13). The most common structures are Aquatic Environments with Vegetation (26 stations, total abundance is 42), Forest Edges (20 stations, total abundance is 31), and Semi-open Woody Structures (17 stations, total abundance is 29). The most frequent combinations are Grasslands × Aquatic habitats with vegetation (18 stations, total abundance is 32), Grasslands × Rocky habitats (13 stations, total abundance is 25), and Grasslands × Forest edges (13 stations, total abundance is 22). No robust effects were found. For *A. sylvaticus*, the results remain descriptives.

4.4. *Chionomys nivalis*

The three most common typologies are Grasslands (36 sites, total abundance is 101), Forest habitats (4 sites, total abundance is 11), and Wetlands (3 sites, total abundance is 5). The three most observed structures are Rocky habitats (18 stations, total abundance is 47), Semi-open woody structures (10 stations, total abundance is 25), and Aquatic habitats with vegetation (9 stations, total abundance is 22). The three most frequent combinations are Grasslands x Rocky habitats (15 stations, total abundance is 40), Grasslands x Semi-open woody structures (10 stations, total abundance is 25), and Grasslands x Aquatic habitats with vegetation (7 stations, total abundance is 17). Wetlands emerged as negatively associated with the species abundance in the harmonized analysis, with an RR = 0.18 (95% CI: 0.05–0.63; qBH = 0.0216). The effect remains detectable after adjustment for BIO2 and BIO12, indicating that this association is not entirely explained by these climatic variables. The second hierarchical level within wetlands did not allow for the identification of any specific fine subcategory responsible for this effect.

4.5. *Clethrionomys glareolus*

The three most present typologies are Grasslands (211 sites, total abundance is 949), Wetlands (44 sites, total abundance is 203), and Forested areas (40 sites, total abundance is 246). The three most identified structures are Forest Edges (108 sites, total abundance is 591), Aquatic Environments with Vegetation (82 sites, total abundance is 457), and Semi-open Woody Structures (74 sites, total abundance is 301). The three most frequent combinations are Grasslands x Forest edges (78 stations, total abundance is 402), Grasslands x Semi-open woody structures (64 stations, total abundance is 261), and Grasslands x Aquatic habitats with vegetation (56 stations, total abundance is 326). Wetlands are negatively associated with abundance (RR = 0.63; 95% CI: 0.45–0.88; qBH = 0.0103), whereas forest habitats are positively associated (RR = 1.90; 95% CI: 1.23–2.93; qBH = 0.0076). Forest edges are also positively associated with abundance (RR = 1.68; 95% CI: 1.26–2.23; qBH = 0.00061). The combination of Grasslands × Forest Edges is also positive (RR = 1.47; 95% CI: 1.06–2.03; qBH = 0.0257). All these results hold after adjustment for bio1, bio2, bio5, bio12, and PCA1, indicating a relationship independent of those variables. The second hierarchical pass did not allow for the identification of specific subcategories for forest habitats or forest edge. The interpretation must therefore remain at the level of the broad categories “Forest Habitats” and “Forest Edges.”

4.6. *Microtus agrestis*

The three most observed typologies are Wetlands (30 sites, total abundance is 104), Grasslands (16 sites, total abundance is 62), and Forested areas (2 sites, total abundance is 5). The three most common structures are Aquatic habitats without vegetation (20 stations, total abundance is 75), Forest edges (16 stations, total abundance is 47), and Semi-open woody structures (8 stations, total abundance is 36). The three most frequent combinations are Wetlands x Aquatic environments without vegetation (13 stations, total abundance is 46), Wetlands x Forest edges (9 stations, total abundance is 30), and Grasslands x Aquatic environments without vegetation (7 stations, total abundance is 29). Wetlands are strongly and positively associated with the species abundance (RR = 6.27; 95% CI: 2.54–15.51; qBH < 0.001), and grasslands are negatively associated (RR = 0.27; 95% CI: 0.12–0.62; qBH = 0.0067). Aquatic habitats without vegetation are positively associated with abundance (RR = 3.96; 95% CI: 1.40–11.20; qBH = 0.0094). At the combined level, Wetlands x Aquatic environments without vegetation are

strongly positive ($RR = 5.61$; 95% CI: 1.15–27.41; $qBH = 0.0329$), and the complementary analysis at the global level also confirms a positive effect of Wetlands x Aquatic habitats ($RR = 4.97$; 95% CI: 1.31–18.9; $qBH = 0.0307$). All these results hold after climate adjustment, indicating that the observed habitat associations are not explained only by BIO1, BIO2, BIO5, BIO12, or PCA1. The second hierarchical analysis did not identify any subcategories of wetlands.

4.7. *Microtus arvalis*

The three most common typologies are Grasslands (26 sites, total abundance is 56), Wetlands (14 sites, total abundance is 43), and Forests (1 site, total abundance is 1). The three most identified structures are Aquatic Environments without Vegetation (16 stations, total abundance is 43), Semi-open Woody Structures (11 stations, total abundance is 22), and Forest edges (9 stations, total abundance is 21). The three most frequent combinations are Grasslands x Aquatic habitats without vegetation (10 stations, total abundance is 20), Grasslands x Semi-open woody structures (8 stations, total abundance is 14), and Grasslands x Forest edges (7 stations, total abundance is 18). Wetlands are positively associated with the species abundance ($RR = 3.05$; 95% CI: 1.18–7.90; $qBH = 0.0263$), and Aquatic habitats without vegetation also show a robust positive effect ($RR = 3.83$; 95% CI: 1.42–10.32; $qBH = 0.0128$). These results hold after adjustment for all tested climatic variables, indicating an independent relationship of the variables BIO1, BIO2, BIO5, BIO12, and PCA1.

The second hierarchical analysis of the structures suggests that the two fine subcategories, Stream (without vegetation) and Torrent (without vegetation), both become visible after adjustment with bio1, bio2, bio5, bio12 and PCA1.

These results indicate a clear association with wetlands and aquatic environments without vegetation.

4.8. *Microtus lavernedii*

The three most identified typologies are Grasslands (21 sites, total abundance is 69), Wetlands (7 sites, total abundance is 22), and Forested habitats (2 sites, total abundance is 2). The three most common structures are Forest edges (11 stations, total abundance is 42), Semi-open Woody Structures (6 stations, total abundance is 26), and Aquatic habitats without Vegetation (5

stations, total abundance is 10). The three most frequent combinations are Grasslands x Forest edges (8 stations, total abundance is 36), Grasslands x Semi-open woody structures (4 stations, total abundance is 13), and Grasslands x Rocky habitats (3 stations, total abundance is 20). At the harmonized level, *Mictodus lavernedii* does not show a robust effect in the raw model. However, several effects become visible after climatic adjustment. Forest edges are positively associated with abundance after adjustment for BIO2 (RR = 4.70; 95% CI: 1.56–14.18; qBH = 0.0365). The combination Grasslands × Forest edges is also positively associated with abundance after adjustment for BIO2 and BIO12 (RR = 7.77; 95% CI: 2.30–26.19; qBH = 0.0057 in the BIO2-adjusted model). On the opposite, Aquatic habitats show a negative association after adjustment for BIO2 and BIO12 (RR = 0.25; 95% CI: 0.08–0.75; qBH = 0.0470 in the BIO2-adjusted model). These effects should therefore be interpreted as visible after adjustment rather than as robust raw-model effects. The second hierarchical pass details some of the results: among the fine subcategories of Forest Eges, Forest edges (coniferous) is showing a robust effect

4.9. *Neomys fodiens*

The three most common typologies are Grasslands (24 sites; total abundance = 34), Wetlands (16 sites, total abundance is 24), and Forest habitats (2 sites, total abundance is 4). The three most observed structures are Forest Edges (18 stations, total abundance is 26), Aquatic Environments without Vegetation (17 stations, total abundance is 27), and Aquatic Environments with Vegetation (10 stations, total abundance is 19). The three mostpresent combinations are Wetlands x Aquatic Habitat (9 stations, total abundance is 14), Grasslands x Aquatic environments without vegetation (9 stations, total abundance is 14), and Wetlands x Forest edges (9 stations, total abundance is 12). Wetlands were positively associated with abundance (RR = 2.36; 95% CI: 1.11–5.06; qBH = 0.0431). Aquatic environments without vegetation also show a robust positive effect (RR = 3.60; 95% CI: 1.70–7.63; qBH = 0.0011). At the combined level, Wetlands x Forests edges is positively associated with abundance (RR = 3.45; 95% CI: 1.12–10.69; qBH = 0.0464). With the dominant humid–aquatic gradient, this combination represents another relevant result. Further analysis at the overall aquatic level shows that the association is even broader: the Aquatic Habitat category too is positively associated with abundance (RR = 3.56; 95% CI: 1.80–7.04; qBH = 0.00034), and the combination of Wetlands x Aquatic Habitat also is seen as robust (RR = 3.61; 95% CI: 1.40–9.30; qBH = 0.0113). On the opposite, semi-open woody structures seem to be negatively

associated with abundance (RR = 0.32; 95% CI: 0.12–0.86; qBH = 0.0300). The second hierarchical analysis did not identify a single distinct subcategory with wetlands or aquatic environments without vegetation. These effects persisted after climate adjustment, indicating a relationship independent of the variables BIO1, BIO2, BIO5, BIO12, and PCA1.

4.10. *Sorex antinorii*

The three most common habitat types are Grasslands (37 sites; total abundance = 63), Wetlands (15 sites; total abundance = 27), and Forests (11 sites; total abundance = 19). The three most common structures are Aquatic habitats with vegetation (21 stations; total abundance = 40), Forest edges (18 stations; total abundance = 35), and Rocky habitats (13 stations; total abundance = 30). The three most observed combinations are Grasslands × Forest edges (14 stations; total abundance = 30), Grasslands × Aquatic habitats with vegetation (13 stations; total abundance = 22), and Grasslands × Rocky habitats (9 stations; total abundance = 17). No robust effects were detected for this species. The results therefore remain descriptive.

4.11. *Sorex araneus*

The three most observed habitat types are Grasslands (75 sites; total abundance = 161), Wetlands (31 sites; total abundance = 64), and Forested areas (7 sites; total abundance = 8). The three most common structures are Forest edges (32 stations; total abundance = 60), Aquatic Environments without Vegetation (26 stations; total abundance = 58), and Aquatic Environments with Vegetation (25 stations; total abundance = 61). The three most frequent combinations are Grasslands x Forest edges (20 stations; total abundance = 40), Grasslands x Aquatic environments with vegetation (17 stations; total abundance = 46), and Grasslands x Aquatic environments without vegetation (17 stations; total abundance = 41). Forest habitats are negatively associated with abundance (RR = 0.34; 95% CI: 0.14–0.86; qBH = 0.0262), with the association remaining significant after adjustment for BIO1, BIO2, BIO5, BIO12, and PCA1. The second hierarchical analysis within forest habitats did not identify any specific subcategory responsible for this association. At the combination level, Grasslands x Semi-open woody structures is also negatively associated with abundance (RR = 0.46; 95% CI: 0.26–0.84; qBH = 0.0191).

4.12. General synthesis

Overall, the results show that habitat associations are strongly species-specific. Wetlands were generally positively associated with the abundance of *M. agrestis*, *M. arvalis* and *N. fodiens*, but negatively associated with *C. nivalis* and *C. glareolus*. Forest-related categories showed contrasting effects, with positive associations for *C. glareolus* and negative associations for *S. araneus*. Aquatic habitats, especially aquatic habitats without vegetation, are mainly associated with *M. agrestis*, *M. arvalis* and *N. fodiens*. Several significant effects also appeared for the combined typology x structure level, including Grasslands x Aquatic habitats with vegetation in *A. alpicola*, Grasslands x Forest edges in *C. glareolus* and *M. lavernedii*, and Wetlands x Aquatic habitats or Wetlands x Forest edges in *N. fodiens* and *M. agrestis*.

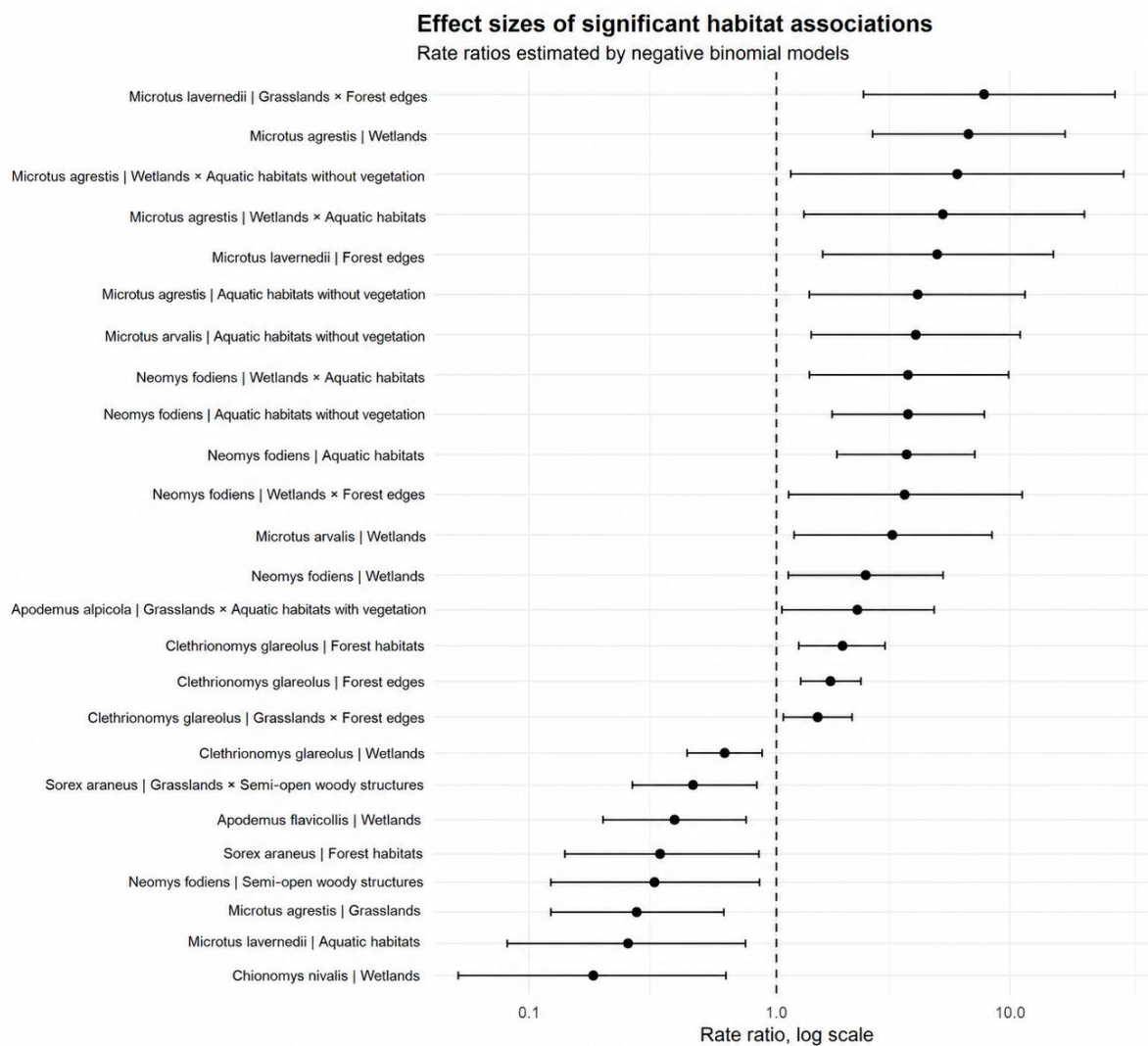


Figure 4. Effect sizes of significant habitat associations. Points represent rate ratios estimated by negative binomial models, and horizontal bars represent 95% confidence intervals. The dashed vertical line indicates no effect.

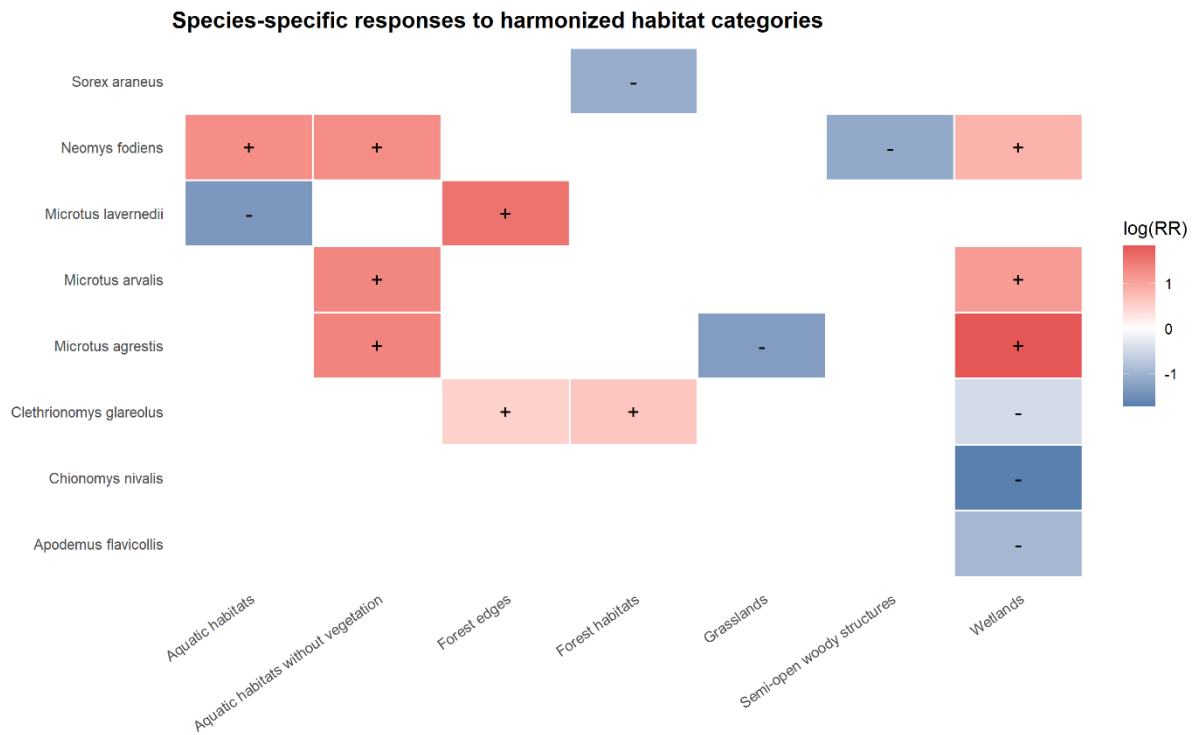


Figure 5. Species-specific responses to harmonized habitat categories. Positive and negative significant associations between species abundance and harmonized habitat categories.

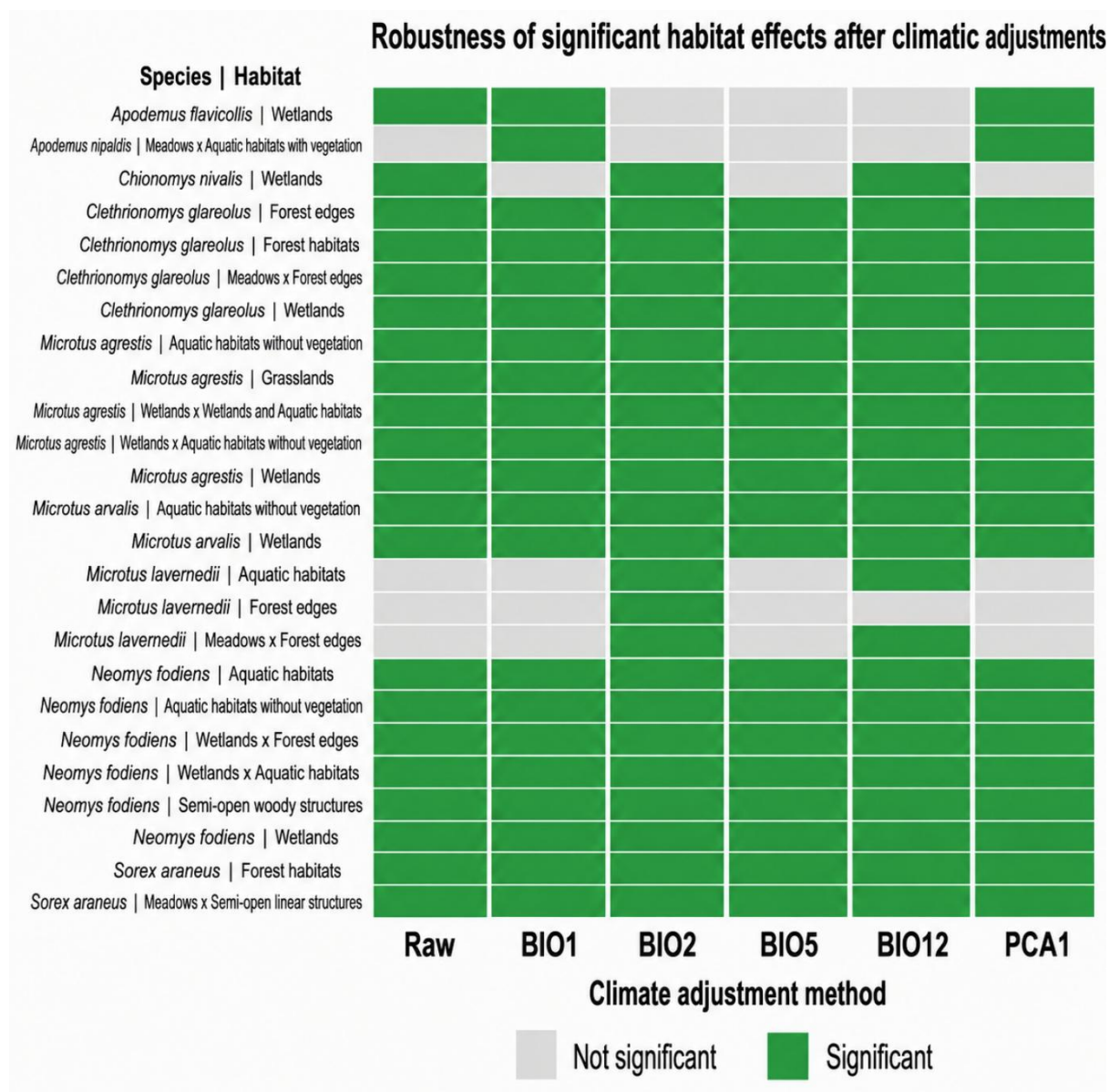


Figure 6. Robustness of significant habitat effects after climatic adjustment. Each row represents a significant habitat effect. Filled cells indicate that the effect remained significant in the raw model or after adjustment for the corresponding climatic variable. Climatic covariates were introduced separately.

5. Discussion

The results show a differentiated answer from micromammals regarding the composition of the mountain agricultural environment. Overall, three main points emerge from our analyses. The first one is the importance of wetlands, that appear to be favourable for some species but disadvantageous for others. The second is the importance of aquatic habitats, especially for aquatic habitats without vegetation, that stand out for multiple species. A third important point is that several significant effects are not linked to typologies or structures considered separately, but to their combination. This is particularly relevant because it shows that some species respond too to local ecological configurations rather than broad habitat categories alone. In this study, combinations such as Grasslands x Aquatic habitats with vegetation, Grasslands x Forest edges, Wetlands x Aquatic habitats, or Wetlands x Forest edges provide additional ecological information and help identify more precise habitat contexts associated with abundance.

5.1. Contrast between wetlands, forests, and grasslands

The most notable result is the character non unanimous of the wetlands. They are associated positively with the abundance of *M. agrestis*, *M. arvalis* and *N. fodiens*, but negatively with *C. nivalis* and *C. glareolus*. Hence, the wetlands can't be considered as globally favourable or unfavourable to all small mammals. Indeed, they favour some species and disadvantage some others.

With *M. agrestis* and *M. arvalis*, the results showing a positive association for abundance with wetlands. This interpretation is strengthened, in the case of *M. agrestis*, by the negative effect on abundance of grassland and by the combinations that highlights the preferences for the aquatic elements (wetlands and aquatic habitats). This suggests that the species is not simply favoured by open habitats, but rather by open habitats that include a wetland or aquatic component. Hence, grassland alone does not appear to provide the same ecological suitability as wetland contexts associated with aquatic structures.

In the *N. fodiens*, the positive effect of wetlands is also observed. The results are not limited to the wetlands and continue with the aquatic structures and with the combinations between wetlands and aquatic structures. The results show that this species depends on wet environments and needs to be close to the water.

On the opposite, the negative associations between wetlands and the abundance of *C. glareolus* and *C. nivalis* suggest that these environments represent less favourable contexts for these species. With those species, this result is particularly convincing because we observed a positive effect of forest habitats and forest edges, and a negative effect on abundance of wetlands. This result shows an opposition between wetlands and related forest habitat.

5.2. The contribution of local structures

The results also show that local structures provide in multiple case valuable ecological information. The typology is not always the only reason explaining the abundance of micromammals species. The locally structure plays an important role too. This point is well illustrated by *A. alpicola*. For this species, no robust effects were detected when typologies or structures were considered separately, whereas the combination of Grasslands x Aquatic habitats with vegetation was positively associated with abundance. This suggests that the species may respond to a mixed ecological configuration, combining an open grassland environment with a locally vegetated aquatic structure. The result supports the relevance of analysing typology and structure combinations, because some habitat associations only become visible at this combined level. In *M. arvalis*, the second hierarchical pass suggests that both streams without vegetation and torrents without vegetation may contribute to the broader effect detected for aquatic habitats without vegetation. However, these fine-level effects appear after climatic adjustment. They shouldn't be interpreted as strong evidence. In *N. fodiens*, the positive association with Wetlands x Forest edges shows that the species is not only linked to strictly aquatic structures. Transitional zones between wetlands and forest habitats may also provide favourable conditions.

For *C. glareolus*, forest edges stand out as a robust element, consistent with the positive effects observed for forest habitats. This finding suggests that the species responds positively not only to presence of a forest environment in the broad sense, but also to the presence of transition area between forest and more open habitats. Forest edges can be favourable for a species with an affinity for forest habitats. The positive association with the combination Grasslands x Forest edges further strengthens this interpretation. It indicates that forest edges may increase the ecological value of grassland dominated environment for *C. glareolus*. Thus, the favourable result is not restricted to forest habitats in the strict sense but also concerns forest edges within open agricultural landscapes.

For *M. agrestis*, *M. arvalis* and *N. fodiens*, a structure is the one of aquatic environment without vegetation. For *Microtus agrestis* and *Microtus arvalis*, this structure reflects a strong affinity for humid environment, combining wetlands and aquatic elements. For *N. fodiens*, the ecological context is clearly aquatic, with a higher abundance in wetlands and aquatic environment, and we can observe an affinity for forest edges.

5.3. What does the second hierarchical pass teach us?

The second hierarchical pass tries to respond to the following question: When a broader harmonized category statistically stands out in the first analysis, is the category related to a particular subcategory? The results obtained show that, in most cases, the answer is negative (Table 2). This shows that the harmonized level corresponds to the level at which the statistical result is the most clearly detectable. For *M. lavernedii*, the main analysis did not reveal a very clear robust effect at the harmonized level. The second pass on the structures show that the effect associated with forest edges is in fact more specifically made by coniferous forest edges. This result shows that the second pass can sometimes provide more ecological precision. A weaker effect also emerges for *M. arvalis* after climatic adjustment, indicating that the second hierarchical level may still provide complementary ecological information.

Species	Analysis level	Parent category	Fine categories with detected effect	Summary
<i>Apodemus flavicollis</i> (Yellow-necked mouse)	Typology second pass	Wetlands	None	No fine-level effect detected
<i>Chionomys nivalis</i>	Typology second pass	Wetlands	None	No fine-level effect detected
<i>Clethrionomys glareolus</i>	Structure second pass	Forest edges	None	No fine-level effect detected
<i>Clethrionomys glareolus</i>	Typology second pass	Forest habitats	None	No fine-level effect detected
<i>Clethrionomys glareolus</i>	Typology second pass	Wetlands	None	No fine-level effect detected
<i>Microtus agrestis</i>	Structure second pass	Aquatic habitats without vegetation	None	No fine-level effect detected
<i>Microtus agrestis</i>	Typology second pass	Wetlands	None	No fine-level effect detected
<i>Microtus arvalis</i>	Structure second pass	Aquatic habitats without vegetation	Stream (without vegetation); Torrent (without vegetation)	Fine-level effect visible only after adjustment
<i>Microtus arvalis</i>	Typology second pass	Wetlands	None	No fine-level effect detected
<i>Microtus lavernedii</i>	Structure second pass	Forest edges	Forest edges (coniferous)	At least one robust fine-level effect
<i>Neomys fodiens</i>	Structure second pass	Aquatic habitats without vegetation	None	No fine-level effect detected

Species	Analysis level	Parent category	Fine categories with detected effect	Summary
<i>Sorex araneus</i>	Typology second pass	Forest habitats	None	No fine-level effect detected

Table 2. Summary of the second hierarchical pass

5.4. Species with no robust signal: no effect or no statistical power?

Several species, such as *A. sylvaticus* and *S. antinorii*, do not show any robust statistical effect. *A. flavicollis* only shows a weak result after climatic adjustment. This absence of result should not be necessarily interpreted as an absence of relation to the environment. It can reflect a statistical power insufficient to detect some differences regarding the typologies or structures. This can come from an insufficient amount of data. *A. flavicollis* shows an intermediate situation. No robust effect is detected in the raw model, which means that the most frequently occupied habitats remain mainly descriptive. However, wetlands show a negative association after adjustment for BIO1 and PCA1. This result should not be interpreted as a robust habitat effect, because it appears only after climatic adjustment. It suggests that part of the preferred habitat may be linked to climatic gradients.

5.5. Limitations of the analysis

Several limitations should be considered in the interpretation of these results. The first one concerns the scarcity of some species or some categories. When numbers become low, especially at the detailed level, the statistical power reduces. This explains why several strong statistically significant results at the harmonized level doesn't hold in the second hierarchical level. Furthermore, as mentioned earlier, some cells in the Excel document remained empty, so I had to enter them manually based on data from the map.geo.admin.ch website and aerial photos, which Increased the risk of error compared to being on-site with experts. The final point, which is also related to the empty cells that had to be entered manually, concerns the temporary discrepancy between the year the small mammals were captured and the year the satellite images were taken. Some captures date back ten years, while the satellite images are more recent. The situation may have changed between those two points in time, but I have no way of knowing.

5.6. Ecological and management implications

Several ecological implications stand out. On one side, the results suggest that the mountain landscapes should not be considered as homogenous from the point of view of the micromammals. Some components of the landscape, in particular the wetlands, forest edges and aquatic structures, play a more important role than others in the organisation of the abundances. On the other side, the varying responses among species show that no single management measure can be considered universally beneficial for the entire community. For example, promoting the intensification of wetlands could advantage some species, such as *M. agrestis*, *M. arvalis* and *N. fodiens*, but could disadvantage other species, such as *C. glareolus* or *C. rivalis*. Similarly, the conservation or maintenance of forest edges appears particularly important for certain species related to forest habitats, but it is not the best solution for the entire species community. Hence, these results therefore support a management approach based on landscape heterogeneity.

5.7. A review of the assumptions made in the introduction

The results make it possible to respond to the three hypotheses formulated in the introduction. Overall, the hypotheses are supported, but not in a uniform way across all species.

The first hypothesis supposed that the wetlands or the aquatic environments, favour particularly the species related to humidity and aquatic environments and are not necessarily favourable to all species. Globally, this hypothesis has been confirmed. The results obtained for *M. agrestis*, *M. arvalis* and *N. fodiens* point in that direction: for those species, wetlands, aquatic structures without vegetation and some combinations between humid typologies and aquatic structures are associated to higher abundance. The case of *N. fodiens* is particularly convincing, because many statistically significant results related to abundance were associated with humid and aquatic contexts. Wetlands are negatively associated with abundance in *C. glareolus* and *C. nivalis*. In addition, *A. flavicollis* shows a weaker negative effect after climatic adjustment. These results indicate that wet and aquatic environments do not act as universally favourable habitats.

The second hypothesis is that the forest habitats and forest edges must advantage forest species and disadvantage some others. The results are as expected, but only partially so. For *C. glareolus*, the results are statistically significant and clear: the forest habitats such as forest edges are positively associated to abundance, and this effect remains unchanged after climate

adjustment. The positive combination between grassland and forest edges strengthens even more this interpretation. *M. lavernedii* provides an interesting result too. Forest edges and the combination Grasslands x Forest edges become significant only after climatic adjustment, and the second hierarchical pass suggests that this response is mainly connected to coniferous forest edges. This result supports the hypothesis only partly, because the result is not robust in the raw model, but it still suggests a possible affinity for structured forest-edge habitats. The second hypothesis is therefore supported for *C. glareolus* and partly for *M. lavernedii*, but not for all species. This hypothesis must also be nuanced by the result obtained for *S. araneus*. In this species, forest habitats are negatively associated with abundance, and this effect remains significant after climatic adjustment. Hence, forest related categories do not have a uniformly positive effect through the community. The hypothesis is strongly supported for *C. glareolus*, partly supported for *M. lavernedii*, but not supported for all species.

The third hypothesis said that the associations between habitat variables and abundance would remain detectable after considering BIO1, BIO2, BIO5, BIO12 and PCA1. Overall, this hypothesis is well supported by our results. Many statistically significant results in the analyses stay unchanged after adjustment. This is the case for the wetland for some species, or the forest habitats and forest edges for *C. glareolus*, or even aquatic structures without vegetation for *M. agrestis*, *M. arvalis* and *N. fodiens*. However, the interpretation should remain cautious for effects that are visible only after adjustment, such as those observed in *A. flavicollis* and *M. lavernedii*. These effects are informative, but they are weaker than effects that are already significant in the raw model and remain significant after adjustment.

6. Conclusion

This study shows that the mountain environment does not influence the abundance of small mammals homogeneously, but the different facets of the mountain habitat greatly influence the presence or absence of different species. The results highlight the important role of several compositions of the landscapes, particularly the wetlands, aquatic structures, forest habitats and forest edges. However, these factors don't affect the entire community in the same way. Firstly, wetlands and aquatic structures seem particularly favourable to several species, in particular *M. agrestis*, *M. arvalis* and *N. fodiens*. On the opposite, they are less favourable, even unfavourable to other species such as *C. glareolus* or *C. nivalis*. Then, forest habitats and forest edges are important for some species in particular for *C. glareolus*, and more cautiously for *M. lavernedii* for the coniferous forest edges. Finally, several results persist after the adjustment of climatic variables, which means that the relations observed are not explained only by a climatic gradient, but also due to the specific influence of local typologies and structures. However, effects that appeared only after climatic adjustment should be interpreted more cautiously, as they are less robust than effects already significant in the raw models. The results also show that some ecological relationships become visible only when typologies and local structures are considered together, as showed by combinations such as Grasslands x Aquatic habitats with vegetation for *Apodemus alpicola*. Other combinations, such as Grasslands x Forest edges, Wetlands x Aquatic habitats and Wetlands x Forest edges, also show that local ecological configurations can provide information that is not visible when typologies or structures are analysed separately.

This master thesis suggests that the results do not point to a single habitat type or structure that should be favoured everywhere, but rather the importance of having diversified landscapes. Wetlands, aquatic structures and forest edges appear to be important components for maintaining a diversity of favourable ecological contexts. However, the contrasted responses observed among species show that no single management recommendation can be considered universally positive for the whole micromammal community.

This study supports the idea that the biodiversity of small mammals in the mountains is influenced by the agricultural environment, but through relationships that depend on the combination of habitat type, local structure, and climatic conditions. This complexity reminds us that a deep comprehension of micromammals requires thinking on multiple levels simultaneously. In this perspective, preserving heterogeneous mountain agricultural landscapes appears essential for maintaining the species and functional diversity of these communities over the long term.

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Appendix

Species	Analysis level	Category	Conclusion	Raw	BIO1	BIO2	BIO5	BIO12	PCA1
<i>Apodemus alpicola</i> (Alpine field mouse)	Typology × structure	Grasslands × Aquatic habitats with vegetation	Robust effect	Yes; RR=2.17; qBH=0.0395	Yes; RR=2.10; qBH=0.0395	Yes; RR=2.17; qBH=0.0395	Yes; RR=2.13; qBH=0.0395	Yes; RR=2.27; qBH=0.0395	Yes; RR=2.24; qBH=0.0395
<i>Apodemus flavicollis</i> (Yellow-necked mouse)	Typology	Wetlands	Visible after adjustment	No	Yes; RR=0.39; qBH=0.0179	No	No	No	Yes; RR=0.39; qBH=0.0179
<i>Chionomys nivalis</i> (Snow vole)	Typology	Wetlands	Robust effect	Yes; RR=0.18; qBH=0.0216	No	Yes; RR=0.17; qBH=0.0216	No	Yes; RR=0.19; qBH=0.0283	No
<i>Clethrionomys glareolus</i> (Bank vole)	Structure	Forest edges	Robust effect	Yes; RR=1.68; qBH=<0.001	Yes; RR=1.65; qBH=<0.001	Yes; RR=1.72; qBH=<0.001	Yes; RR=1.65; qBH=<0.001	Yes; RR=1.70; qBH=<0.001	Yes; RR=1.65; qBH=<0.001
<i>Clethrionomys glareolus</i> (Bank vole)	Structure, overall aquatic level	Forest edges	Robust effect	Yes; RR=1.68; qBH=<0.001	Yes; RR=1.65; qBH=<0.001	Yes; RR=1.72; qBH=<0.001	Yes; RR=1.65; qBH=<0.001	Yes; RR=1.70; qBH=<0.001	Yes; RR=1.65; qBH=<0.001
<i>Clethrionomys glareolus</i> (Bank vole)	Typology	Forest habitats	Robust effect	Yes; RR=1.90; qBH=0.0076	Yes; RR=1.85; qBH=0.0076	Yes; RR=1.87; qBH=0.0076	Yes; RR=1.79; qBH=0.0089	Yes; RR=1.87; qBH=0.0076	Yes; RR=1.84; qBH=0.0076
<i>Clethrionomys glareolus</i> (Bank vole)	Typology	Wetlands	Robust effect	Yes; RR=0.63; qBH=0.0103	Yes; RR=0.57; qBH=0.0032	Yes; RR=0.65; qBH=0.0147	Yes; RR=0.58; qBH=0.0032	Yes; RR=0.66; qBH=0.0212	Yes; RR=0.57; qBH=0.0032
<i>Clethrionomys glareolus</i> (Bank vole)	Typology × structure	Grasslands × Forest edges	Robust effect	Yes; RR=1.47; qBH=0.0257	Yes; RR=1.47; qBH=0.0257	Yes; RR=1.46; qBH=0.0257	Yes; RR=1.47; qBH=0.0257	Yes; RR=1.44; qBH=0.0264	Yes; RR=1.48; qBH=0.0257
<i>Clethrionomys glareolus</i> (Bank vole)	Typology × structure, overall aquatic level	Grasslands × Forest edges	Robust effect	Yes; RR=1.47; qBH=0.0257	Yes; RR=1.47; qBH=0.0257	Yes; RR=1.46; qBH=0.0257	Yes; RR=1.47; qBH=0.0257	Yes; RR=1.44; qBH=0.0264	Yes; RR=1.48; qBH=0.0257
<i>Microtus agrestis</i> (Field vole)	Structure	Aquatic habitats without vegetation	Robust effect	Yes; RR=3.96; qBH=0.0094	Yes; RR=4.55; qBH=0.0031	Yes; RR=4.36; qBH=0.0074	Yes; RR=4.58; qBH=0.0031	Yes; RR=3.58; qBH=0.0075	Yes; RR=4.47; qBH=0.0031
<i>Microtus agrestis</i> (Field vole)	Typology	Wetlands	Robust effect	Yes; RR=6.27; qBH=<0.001	Yes; RR=4.58; qBH=<0.001	Yes; RR=5.81; qBH=<0.001	Yes; RR=4.89; qBH=<0.001	Yes; RR=6.23; qBH=<0.001	Yes; RR=5.10; qBH=<0.001
<i>Microtus agrestis</i> (Field vole)	Typology	Grasslands	Robust effect	Yes; RR=0.27; qBH=0.0067	Yes; RR=0.37; qBH=0.0145	Yes; RR=0.29; qBH=0.0072	Yes; RR=0.35; qBH=0.0113	Yes; RR=0.28; qBH=0.0059	Yes; RR=0.34; qBH=0.0105

Species	Analysis level	Category	Conclusion	Raw	BIO1	BIO2	BIO5	BIO12	PCA1
<i>Microtus agrestis</i> (Field vole)	Typology × structure	Wetlands × Aquatic habitats without vegetation	Robust effect	Yes; RR=5.61; qBH=0.0329	Yes; RR=4.66; qBH=0.0318	Yes; RR=6.19; qBH=0.0318	Yes; RR=5.51; qBH=0.0318	Yes; RR=4.33; qBH=0.0329	Yes; RR=4.96; qBH=0.0318
<i>Microtus agrestis</i> (Field vole)	Typology × structure, overall aquatic level	Wetlands × Aquatic habitats	Robust effect	Yes; RR=4.97; qBH=0.0307	Yes; RR=3.54; qBH=0.0319	Yes; RR=5.58; qBH=0.0307	Yes; RR=4.23; qBH=0.0307	Yes; RR=3.51; qBH=0.0319	Yes; RR=3.72; qBH=0.0307
<i>Microtus arvalis</i> (Common vole)	Structure	Aquatic habitats without vegetation	Robust effect	Yes; RR=3.83; qBH=0.0128	Yes; RR=3.57; qBH=0.0128	Yes; RR=3.59; qBH=0.0128	Yes; RR=3.53; qBH=0.0128	Yes; RR=3.52; qBH=0.0128	Yes; RR=3.63; qBH=0.0128
<i>Microtus arvalis</i> (Common vole)	Typology	Wetlands	Robust effect	Yes; RR=3.05; qBH=0.0263	Yes; RR=4.10; qBH=0.0078	Yes; RR=3.83; qBH=0.0078	Yes; RR=4.61; qBH=0.0052	Yes; RR=2.74; qBH=0.0465	Yes; RR=4.86; qBH=0.0052
<i>Microtus lavernedii</i> (Lavernede's vole)	Structure	Forest edges	Visible after adjustment	No	No	Yes; RR=4.70; qBH=0.0365	No	No	No
<i>Microtus lavernedii</i> (Lavernede's vole)	Structure, overall aquatic level	Forest edges	Visible after adjustment	No	No	Yes; RR=4.70; qBH=0.0365	No	No	No
<i>Microtus lavernedii</i> (Lavernede's vole)	Structure, overall aquatic level	Aquatic habitats	Visible after adjustment	No	No	Yes; RR=0.25; qBH=0.0470	No	Yes; RR=0.26; qBH=0.0470	No
<i>Microtus lavernedii</i> (Lavernede's vole)	Typology × structure	Grasslands × Forest edges	Visible after adjustment	No	No	Yes; RR=7.77; qBH=0.0057	No	Yes; RR=6.01; qBH=0.0097	No
<i>Microtus lavernedii</i> (Lavernede's vole)	Typology × structure, overall aquatic level	Grasslands × Forest edges	Visible after adjustment	No	No	Yes; RR=7.77; qBH=0.0057	No	Yes; RR=6.01; qBH=0.0097	No
<i>Neomys fodiens</i> (Eurasian water shrew)	Structure	Aquatic habitats without vegetation	Robust effect	Yes; RR=3.60; qBH=0.0011	Yes; RR=3.57; qBH=0.0011	Yes; RR=3.42; qBH=0.0012	Yes; RR=3.59; qBH=0.0011	Yes; RR=3.59; qBH=0.0011	Yes; RR=3.58; qBH=0.0011
<i>Neomys fodiens</i> (Eurasian water shrew)	Structure, overall aquatic level	Aquatic habitats	Robust effect	Yes; RR=3.56; qBH=<0.001	Yes; RR=3.53; qBH=<0.001	Yes; RR=3.57; qBH=<0.001	Yes; RR=3.57; qBH=<0.001	Yes; RR=3.50; qBH=<0.001	Yes; RR=3.56; qBH=<0.001
<i>Neomys fodiens</i> (Eurasian water shrew)	Structure, overall aquatic level	Semi-open woody structures	Robust effect	Yes; RR=0.32; qBH=0.0300	Yes; RR=0.33; qBH=0.0300	Yes; RR=0.32; qBH=0.0300	Yes; RR=0.32; qBH=0.0300	Yes; RR=0.33; qBH=0.0300	Yes; RR=0.33; qBH=0.0300
<i>Neomys fodiens</i> (Eurasian water shrew)	Typology	Wetlands	Robust effect	Yes; RR=2.36; qBH=0.0431	Yes; RR=2.32; qBH=0.0449	Yes; RR=2.50; qBH=0.0431	Yes; RR=2.52; qBH=0.0431	Yes; RR=2.28; qBH=0.0449	Yes; RR=2.43; qBH=0.0431

Species	Analysis level	Category	Conclusion	Raw	BIO1	BIO2	BIO5	BIO12	PCA1
Neomys fodiens (Eurasian water shrew)	Typology × structure	Wetlands × Forest edges	Robust effect	Yes; RR=3.45; qBH=0.0464	Yes; RR=3.32; qBH=0.0464	Yes; RR=3.14; qBH=0.0464	Yes; RR=3.46; qBH=0.0464	Yes; RR=3.27; qBH=0.0464	Yes; RR=3.44; qBH=0.0464
Neomys fodiens (Eurasian water shrew)	Typology × structure, overall aquatic level	Wetlands × Aquatic habitats	Robust effect	Yes; RR=3.61; qBH=0.0113	Yes; RR=3.56; qBH=0.0113	Yes; RR=4.30; qBH=0.0113	Yes; RR=4.00; qBH=0.0113	Yes; RR=3.51; qBH=0.0113	Yes; RR=3.79; qBH=0.0113
Sorex araneus (Common shrew)	Typology	Forest habitats	Robust effect	Yes; RR=0.34; qBH=0.0262	Yes; RR=0.33; qBH=0.0262	Yes; RR=0.35; qBH=0.0262	Yes; RR=0.35; qBH=0.0262	Yes; RR=0.34; qBH=0.0262	Yes; RR=0.32; qBH=0.0262
Sorex araneus (Common shrew)	Typology × structure, overall aquatic level	Grasslands × Semi-open woody structures	Robust effect	Yes; RR=0.46; qBH=0.0191	Yes; RR=0.47; qBH=0.0191	Yes; RR=0.47; qBH=0.0191	Yes; RR=0.43; qBH=0.0191	Yes; RR=0.52; qBH=0.0291	Yes; RR=0.48; qBH=0.0191

Table 3. Climate-adjusted robustness of significant effects

Declaration on generative AI

During the preparation of this thesis, I used ChatGPT (OpenAI) to assist with R coding, including code development, debugging, and clarification of programming procedures. All statistical analyses and figures were performed in R. I reviewed and verified all AI-assisted content and take full responsibility for the final content of this thesis.

The data supporting the results and the code are archived on GitHub (https://github.com/LEF-UNINE/Peterschmitt_Milan_MSc_thesis_2026).